

Skill Check 3: Building Better Code

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Asking for help on specific questions. (It is fine to ask for clarification!)
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully. For each two questions correct, you will receive one course point for a total of two course points.

Question 1: FUNCTIONS

Write a function that calculates the area of an octagon. The equation for the area of an octagon is $area = 2(1 + \sqrt{2}) * s^2$, where s is the length of one side. Include an example function call to demonstrate the function in operation.

[illegible]

Question 2: LOOPS

Create a while or repeat loop that runs a counter and prints each value of the counter until it reaches 25.

[illegible]

Question 3: CONDITIONALS

Load in the mtcars dataset. Write a conditional that prints “Long dataset!” if the dataset has more than 100 rows, otherwise, print “Small dataset.”

[illegible]

Question 4: COMBINATIONS

You are testing the effectiveness of a drug on individual trial participants, and the data are contained in a data frame named **Theoph**. Each subject receives a different dose and is a different weight, and you want to compare the concentration in their urine over time with the dose per unit weight they received at the beginning of the study. These are the first three calculations:

```
dosewt_sub1 <- Theoph$Dose[Theoph$Subject == 1][1] /  
               Theoph$Wt[Theoph$Subject == 1][1]  
dosewt_sub2 <- Theoph$Dose[Theoph$Subject == 2][1] /  
               Theoph$Wt[Theoph$Subject == 2][1]  
dosewt_sub3 <- Theoph$Dose[Theoph$Subject == 3][1] /  
               Theoph$Wt[Theoph$Subject == 3][1]
```

There are 12 subjects in the study. Create a loop that calculates all the dosewt values for each subject and stores the results in a vector object **dosewts**.

[illegible]